
AI-Driven Churn Prediction & Retention Strategy

A Data-Driven Business Proposal for Company A

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Current State of the Telecom Industry

Wireless operators face intensifying competition and rising customer churn.



Acquiring a new customer costs 5–10× more than retaining an existing one.



Typical monthly churn rate for wireless operators in competitive markets.



Industry ARPU is declining, making retention of existing high-value subscribers critical.



The Opportunity for Company A

Company A has accumulated extensive customer data (usage, billing, service quality, handset info) but lacks an in-house analytics capability to predict and prevent churn. **By building an AI churn-risk model, we can identify at-risk customers early enough for targeted retention, reducing revenue loss and improving campaign ROI.**

Prediction Using Machine Learning Models

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Introduction of Data

Company A provided historical customer data across two linked tables totaling ~100,000 records.



Client.csv

100,000 rows × 50 columns

Account demographics, handset info, household attributes, credit class, geographic area



Record.csv

100,000 rows × 51 columns

Usage, billing, call quality, dropped/blocked calls, overage, roaming, churn label

Joined on Customer_ID (1:1 inner merge)

Dataset at a Glance

Total Customers

100,000

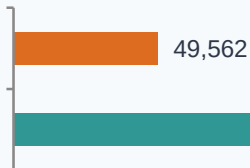
Total Features

~100 columns

Target Variable

churn (31–60 days)

Churned (49,562)



49,562

Stayed (50,438)

50,438

Data Visualization

We conducted exploratory data analysis across 100,000 customers to identify the key churn patterns.



Device Age Is the #1 Separator

Equipment age (eqpdays) has the strongest effect (Cohen's $d = 0.23$). Churn rises from 40.7% for newest devices to 55.7% for oldest. Cheap handsets also increase risk.



Usage Decline Signals Disengagement

Churners average 483 minutes vs 543 for non-churners. Recent usage change (change_mou) is more predictive than demographics. Trend features outperform static attributes.



Surprise: Support Calls $\neq$ Churn Risk

The naive "more complaints = more churn" hypothesis is contradicted. Churners actually make fewer customer-care calls on average (1.60 vs 1.98). Support contact alone is not a risk signal.

Churn Risk Pockets

When features combine, risk amplifies:

Feature Combination	Churn %
Low usage + High bill	58.6%
Cheap device + Low usage	59.6%
Low usage + High drops	63.7%
High usage + High bill	42.3%
Premium device + High usage	38.8%

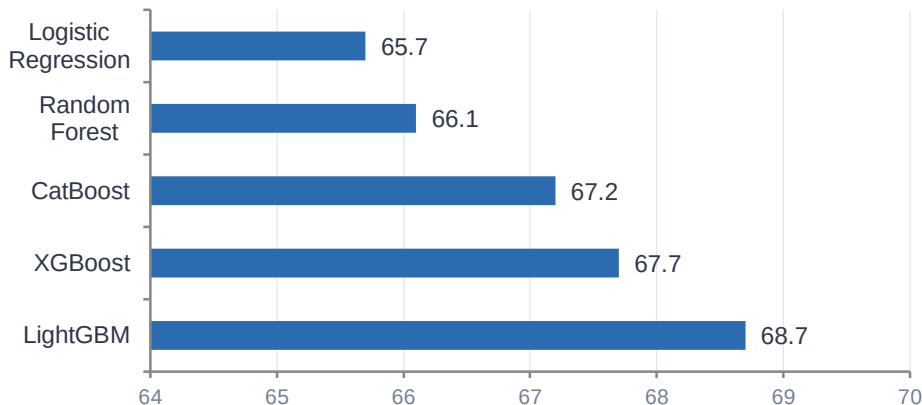
Individual features are weak separators. Combined risk pockets reach 60%+ churn, making multivariate models essential.

Overview of the Machine Learning Model

From customer behavior, device, and service data, we predict the probability of churn within 31–60 days.



5-Model Baseline Comparison — ROC-AUC × 100 (5-Fold CV)



LightGBM selected as the best-performing model with the lowest generalization gap.



Final Tuned Model

LightGBM + Optuna (100 trials)

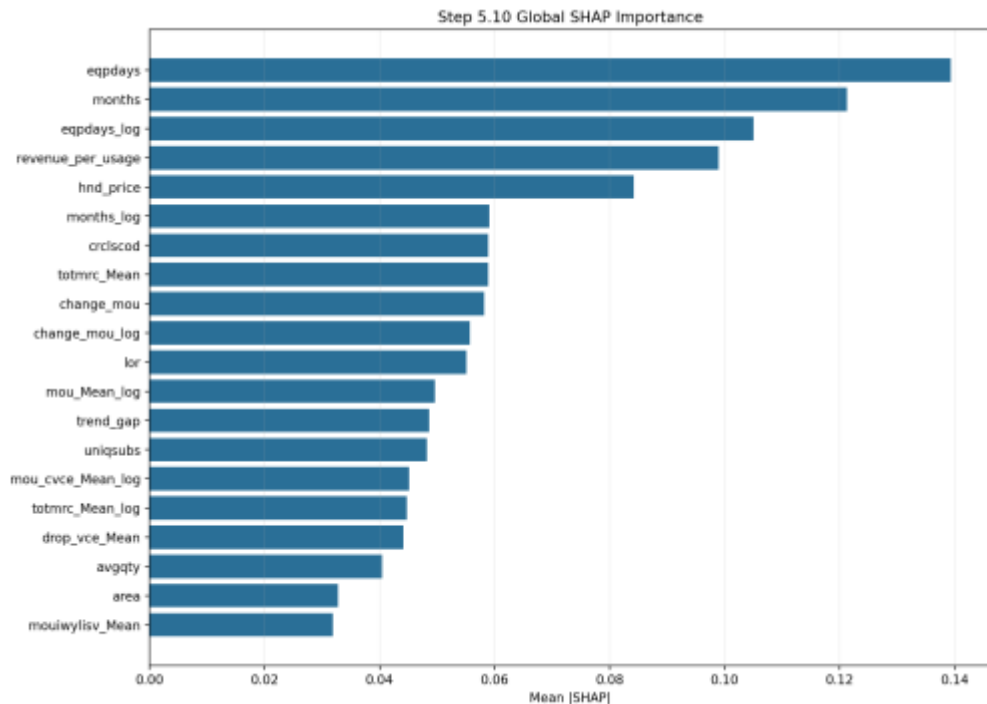
ROC-AUC	0.695
Accuracy	63.9%
Precision	63.3%
Recall	64.5%
F1 Score	63.9%

Evaluation

5-Fold Stratified CV

Important Features for Prediction

SHAP analysis reveals how each feature contributes to individual churn predictions.



Top Churn Drivers

months	Short tenure → high risk	0.121
hnd_price	Cheap handset → high risk	0.084
totmrc_Mean	Low recurring charge	0.059
change_mou	Usage decline	0.058
drop_vce_Mean	Dropped calls	0.044

Top Retention Drivers

eqpdays	Newer equipment	0.139
crclscod	Better credit class	0.059
uniqsubs	Larger household	0.048

The model is nonlinear and segmented — device lifecycle, tenure, and usage trends dominate over demographics. See dependence plots for detailed feature behavior.

Our Business Proposal

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Our Business Proposals

Two data-driven proposals to reduce avoidable churn and protect Company A's recurring revenue.



Proposal 1

AI Churn Risk Scoring

Deploy the LightGBM model to score every customer monthly on churn probability. Retention teams receive a ranked risk list instead of treating all customers equally.

Target: Focus retention budget on the top 20% highest-risk customers



Proposal 2

Segment-Based Proactive Retention

Use SHAP-identified risk segments to design specific interventions: handset upgrade offers for old-device customers, plan adjustments for high-bill/low-usage users, and service follow-ups for drop-call victims.

Target: Match the right retention action to each risk profile

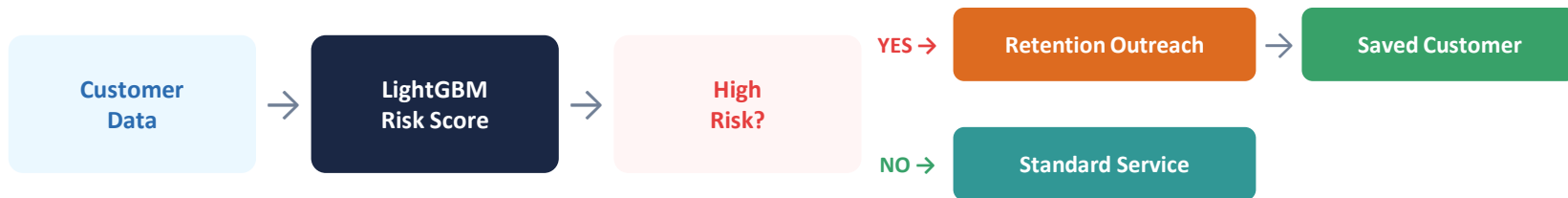


Long-Term Support

Model retraining on fresh data quarterly • Seasonal churn pattern analysis • Dashboard integration for retention teams • A/B testing retention offer effectiveness

Proposal 1: AI Churn Risk Scoring

Score every customer monthly and route high-risk accounts to the retention team.



Overview

Score every customer monthly on churn probability using the tuned LightGBM model (ROC-AUC 0.695).

The model uses 61 features covering device age, usage trends, billing, and service quality to produce a 0–100 risk score.

Novelty

Replaces broad, untargeted retention campaigns with data-driven prioritization.

Current approach: spend equally on all customers. New approach: concentrate 80% of budget on the top 20% risk customers where save probability is highest.

Expected Effects

For customers: proactive service before frustration leads to churn.

For Company A: focused spend on high-risk accounts, higher save rate, lower cost-per-save, and improved campaign ROI.

Validity of the Churn Risk Score

SHAP dependence plots confirm the model captures meaningful, nonlinear customer behavior patterns.

ROC-AUC

0.695

5-Fold Stratified CV

Accuracy

63.9%

validation

Precision

63.3%

validation

Recall

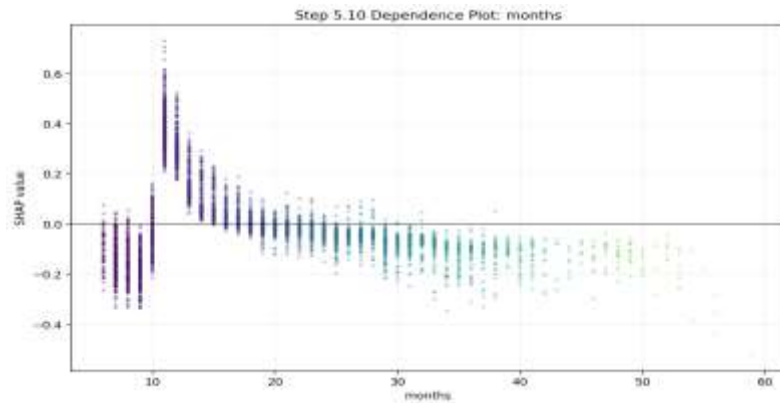
64.5%

validation

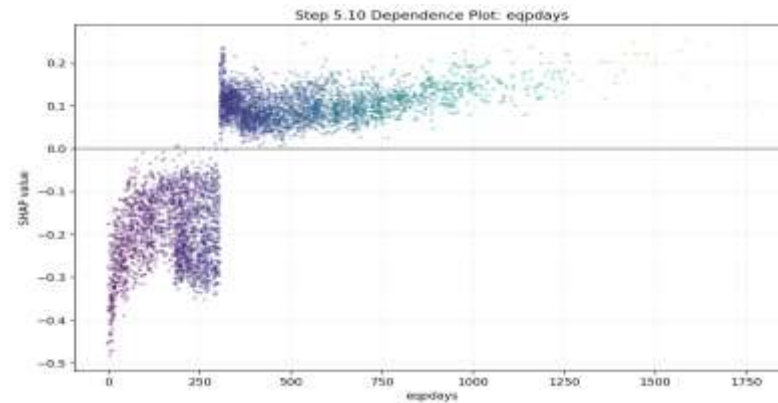
F1 Score

63.9%

validation



Short tenure (6–12 mo) = highest churn risk zone



Newer equipment (<300 days) = strong retention signal

Revenue Impact Quantification

Estimating the financial benefit of AI-driven targeted retention versus untargeted campaigns.

Assumptions

Parameter	Value
Total customers	100,000
Observed churn rate	49.6%
Est. monthly ARPU	\$55
Annual revenue per customer	\$660
Model targets top-risk %	Top 20%
Customers targeted	~9,912
Save rate (targeted intervention)	15%
Cost per intervention	\$30

Based on rev_Mean ≈ \$58, industry save rates 10–20%, and standard telecom retention offer costs.

Projected Annual Impact

Customers Saved ~1,487

Revenue Protected \$981K

Intervention Cost **-\$297K**

Net Annual Benefit \$684K

ROI: ~230% return on retention investment

Proposal 2: Segment-Based Proactive Retention

SHAP analysis identifies three actionable customer risk segments, each with a tailored intervention.



Segment A

Old Device, Low Engagement

Trigger

eqpdays > 400 days + declining mou_Mean

Action

Subsidized handset upgrade offer tied to contract renewal

Expected Outcome

Re-engage customer with new device lock-in; reset eqpdays to 0



Segment B

High Bill, Low Usage

Trigger

High totmrc_Mean + low mou_Mean (bill-to-usage mismatch)

Action

Plan right-sizing or loyalty discount to reduce bill shock

Expected Outcome

Reduce churn from 58.6% risk pocket; improve perceived value



Segment C

Service Quality Issues

Trigger

Elevated drop_vce_Mean + declining change_mou

Action

Priority service follow-up + network team escalation in affected area

Expected Outcome

Fix root cause before customer leaves; reduce 63.7% risk pocket

Implementation Roadmap

A phased rollout to minimize risk and maximize learning at each stage.



Commitment: We provide end-to-end support from data integration to model monitoring. Company A's retention team receives actionable risk lists — no ML expertise required.

Summary

Company A can reduce annual churn-related revenue loss by ~\$684K through AI-driven risk scoring and targeted, segment-specific retention actions.

Model Performance

ROC-AUC 0.695

LightGBM + Optuna tuned

Feature Engineering

61 Features

from 227 → compact & interpretable

Net Annual Benefit

\$684K

~230% ROI on retention spend

Thank You